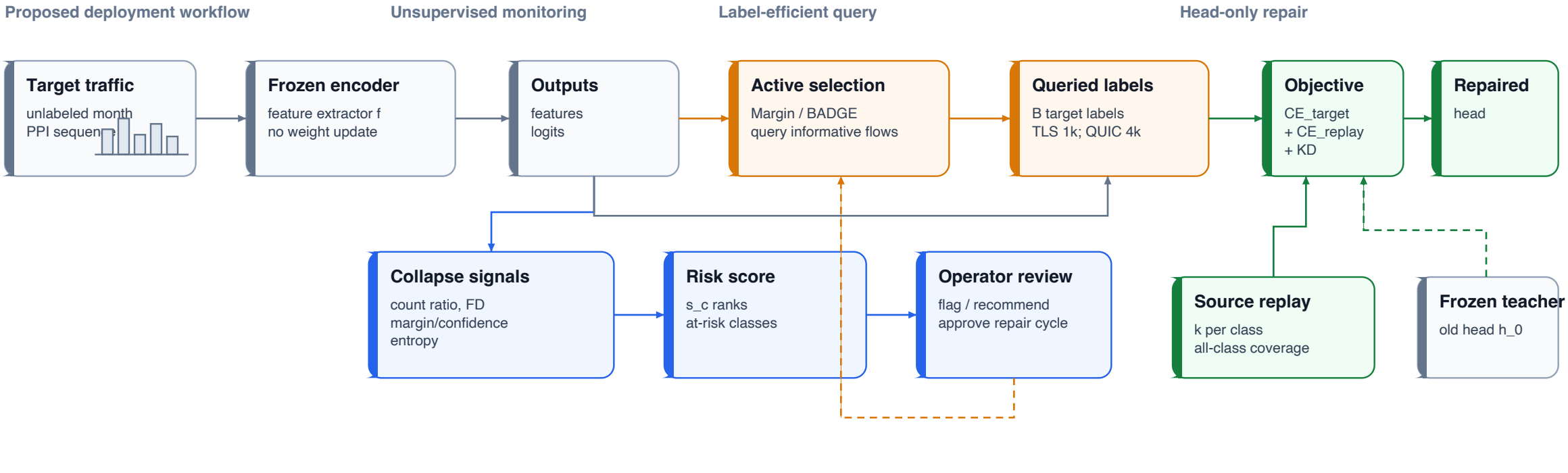
Proposed deployment workflow
Unsupervised monitoring
Label-efficient query
Head-only repair
Target traffic
unlabeled month
PPI sequence
Frozen encoder
feature extractor f
no weight update
Outputs
features
logits
Active selection
Margin / BADGE
query informative flows
Queried labels
B target labels
TLS 1k; QUIC 4k
Objective
CE_target
+ CE_replay
+ KD
Repaired
head
Collapse signals
count ratio, FD
margin/confidence
entropy
Risk score
s_c ranks
at-risk classes
Operator review
flag / recommend
approve repair cycle
Source replay
k per class
all-class coverage
Frozen teacher
old head h_0
solid: data/features
dashed: approval / KD
Encoder stays frozen; only the classification head is updated.